

Quiz 1

1) Given O.D.E is

$$x dy + (y - 2y^2 x^3) dx = 0 \text{ ————— } (*)$$

$$\Rightarrow x \frac{dy}{dx} + y = 2y^2 x^3$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = 2y^2 x^2$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = 2x^2 \text{ ————— } (i)$$

$$\text{Let } z = \frac{1}{y} \Rightarrow \frac{dz}{dx} = -\frac{1}{y^2} \frac{dy}{dx}$$

$\therefore$ Eqⁿ (i) becomes

$$\frac{dz}{dx} - \frac{z}{x} = -2x^2$$

which is required Bernoulli equation form
of (*)

$$\text{I.F} = e^{\int -\frac{1}{x} dx} = e^{\log \frac{1}{x}} = \frac{1}{x}$$

Therefore, solⁿ is

$$z \cdot \frac{1}{x} = \int \frac{1}{x} \cdot -2x^2 dx + C$$

$$\Rightarrow \frac{z}{x} = -x^2 + C$$

$$\Rightarrow z = -x^3 + Cx$$

$$\Rightarrow \frac{1}{y} = -x^3 + Cx$$

i.e. $y = \frac{1}{Cx - x^3}$ is required solⁿ of $\textcircled{*}$. $\square$

2) let $y=f(x)$ be curve (mirror) and $P(x, y)$ be
 $\textcircled{a}$ a point on the curve.

let OP be an incident ray and PC be the reflected ray.

Since angle of incidence is equal to angle of reflection therefore we have

$$\angle APD = \angle BPC = \beta$$

Also the lines AD and PC are parallel, we have
 $\phi = \beta$.

From triangle $\triangle ODP$, we get

$$\theta + (\pi - \beta) + \angle ODP = 2\pi$$

$$\text{i.e. } \angle ODP = \pi - \theta + \beta$$

Now, from $\triangle ADP$, we get

$$\phi + \pi + \angle ADP = 2\pi$$

$$\text{or } \phi + \angle ODP = \pi$$

$$\text{i.e. } \phi + \pi - \theta + \beta = \pi$$

$$\text{i.e. } \phi - \theta + \beta = 0$$

$$\text{i.e. } \beta - \theta + \beta = 0 \quad \text{i.e. } \theta = 2\beta$$

Note that, $\tan \theta = y/x$. And the slope of the tangent to the curved mirror at the point P is given to be $\tan \phi = \frac{dy}{dx}$.

$$\text{Now, } \tan \theta = \tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

$$\text{Then, } \frac{y}{x} = \frac{2y'}{1 - (y')^2}$$

$$\Rightarrow y(y')^2 + 2y'x - y = 0$$

$$\therefore y' = \frac{-2x \pm \sqrt{4x^2 + 4y^2}}{2y}$$

$$= \frac{-x \pm \sqrt{x^2 + y^2}}{y}$$

$$\Rightarrow y dy + x dx = \pm \sqrt{x^2 + y^2} dx$$

$$\Rightarrow \frac{y dy + x dx}{\sqrt{y^2 + x^2}} = \pm dx$$

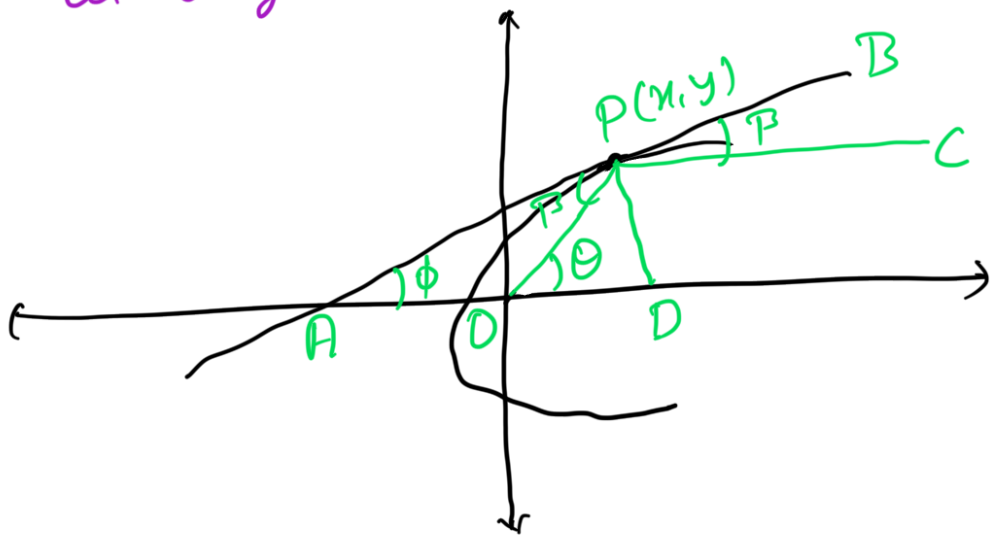
$$\Rightarrow \int d(\sqrt{y^2 + x^2}) = \pm \int dx + C$$

$$\sqrt{y^2 + x^2} = \pm x + C$$

$$\Rightarrow y^2 + x^2 = \pm(x + c)$$

$$\Rightarrow y^2 + x^2 = (x + c)^2$$

i.e. $y^2 = 2xc + c^2$, which is required family of parabolas with focus at origin & axis along the x-axis. $\square$



⑥ Given differential Equation is $(x + ye^{2xy})dx + nx e^{2xy}dy = 0$ $\text{---} (*)$

Here, $M = x + ye^{2xy}$, $N = nx e^{2xy}$

For exact D.E we must have

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\text{i.e. } e^{2xy} + 2xye^{2xy} = ne^{2xy} + 2xyn e^{2xy}$$

By comparing the coefficient of e^{2xy} we

$$\text{have } n = 1$$

∴ we have

Putting $n=1$ in $(*)$ we get

$$(x + y e^{2xy}) dx + x e^{2xy} dy = 0 \quad \text{--- (1)}$$

Since (1) is exact this will imply
 $\exists u(x, y) = b$ s.t. $du = M dx + N dy$.

$$u(x, y) = \int M dx + f(y) ; f(y) \text{ is arbitrary f'n of } y.$$

$$\Rightarrow u(x, y) = \int (x + y e^{2xy}) dx + f(y)$$

$$\Rightarrow u(x, y) = \frac{x^2}{2} + \frac{y e^{2xy}}{2y} + f(y)$$

Differentiating partially w.r.t y

$$\frac{\partial u}{\partial y} = \frac{2x e^{2xy}}{2} + f'(y)$$

$$\text{Since } \frac{\partial u}{\partial y} = N$$

$\therefore$ we have,

$$x e^{2xy} = x e^{2xy} + f'(y)$$

$$\text{i.e. } f'(y) = 0 \Rightarrow f(y) = C$$

Hence, the soln is

$$u(x, y) = \frac{1}{2} (x^2 + e^{2xy}) + C$$

$$\text{i.e. } x^2 + e^{2xy} = 2(b - C)$$

$$\text{i.e. } x^2 + e^{2xy} = C_1, \quad C_1 \text{ is arbitrary const.}$$

is required solⁿ of ①.

□

③ Given family of Curve is

$$x^2 + y^2 - 2Cx = 0 \quad \text{--- ①}$$

Differentiating both sides w.r.t ①

$$x + y \frac{dy}{dx} = C$$

$$\Rightarrow \frac{dy}{dx} = \frac{C-x}{y} \quad \text{--- ②}$$

From ① we $C = \frac{x^2 + y^2}{2x}$

$\therefore$ ② becomes

$$\frac{dy}{dx} = \frac{\frac{x^2 + y^2}{2x} - x}{y} = \frac{y^2 - x^2}{2xy}$$

The orthogonal trajectory satisfy

$$\frac{dy}{dx} = -\frac{2xy}{y^2 - x^2}$$

let $y = vx$ then $\frac{dy}{dx} = v + x \frac{dv}{dx}$

Thus, $v + x \frac{dv}{dx} = \frac{2v}{1-v^2}$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v}{1-v^2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v - v + v^3}{1-v^2} = \frac{v+v^3}{1-v^2}$$

$$\Rightarrow \frac{1-v^2}{v+v^3} dv = \frac{dx}{x}$$

$$\Rightarrow \int \frac{dv}{v} - \int \frac{2v}{1+v^2} dv = \int \frac{dx}{x} + \ln C$$

$$\Rightarrow \ln v - \ln(1+v^2) = \ln x + \ln C$$

$$\Rightarrow \frac{v}{1+v^2} = xC$$

$$\Rightarrow \frac{y/x}{1+y^2/x^2} = xC$$

$$\Rightarrow y = C(x^2+y^2) \text{ is required family. } \square$$

up a/b Given IVP is

$$xy' - y = 0, y(0) = 0 \text{ ——— ①}$$

$$\Rightarrow y' = f(x, y) = y/x$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{x} \Rightarrow \int \frac{dy}{y} = \int \frac{dx}{x} + \ln C$$

$\Rightarrow y = xC$
 since $y(0) = 0 \Rightarrow C$ have uncountable choices.

i.e. $y = Cx : C \in \mathbb{R}$ is required solⁿ.

Since $f(x, y) = y/x$ is not continuous on domain (rectangle) containing origin therefore hypothesis of

Picard's Thm. does not hold.

Thus, it won't contradict the uniqueness Theorem. $\square$

b) Given IVP is

$$y'' - k^2 y = 0 \quad ; \quad k \neq 0$$

$$y(0) = 1, \quad y'(0) = 1$$

$$\text{A.E. is } m^2 - k^2 = 0 \Rightarrow m = \pm k$$

$$\therefore y = C_1 e^{-kx} + C_2 e^{kx} \text{ is solⁿ}. \quad \Rightarrow C_1 = 1 - C_2$$

$$y(0) = 1 = C_1 + C_2 \Rightarrow C_1 = 1 - C_2$$

$$y'(0) = -kC_1 + C_2 k = 1$$

$$\therefore -k + kC_2 + C_2 k = 1$$

$$\Rightarrow C_2 = \frac{1+k}{2k}$$

and $G = \frac{K-1}{2K}$

$$\therefore y = \left(\frac{K-1}{2K}\right)e^{-Kx} + \left(\frac{1+K}{2K}\right)e^{Kx}$$

is particular soln.

